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Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_8_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 8

Reinforcement Learning — Single-Agent Q-Learning and Independent Multi-Agent Learners

Aim

To train a single agent by Q-learning in a grid world whose transitions and rewards are not known in advance, and then to place two independent Q-learners in one cooperative game and observe what changes when the reward of each depends on what the other does at the same moment.

Theory

Reinforcement learning applies where the agent is given no model of its environment: it must estimate the value of acting from the outcomes it experiences. Q-learning keeps one estimate Q(s, a) of the expected discounted return of taking action a in state s and thereafter acting best, and after every transition it moves that estimate towards the reward just received plus the discounted value of the best action available in the new state. The method is off-policy — the update uses the best next action rather than the one actually taken next — so it converges to the optimal values whatever policy generates the experience, provided every state-action pair keeps being tried. That proviso is the whole of the exploration problem, and it is met either by acting randomly a fraction of the time or by initialising the estimates high enough that untried actions look attractive.

The guarantee fails as soon as a second learner is added. Each agent sees only its own action and the reward that follows, so the other agent is simply part of its environment — but that part is itself learning, so the rewards associated with an action keep changing and the environment is no longer stationary. In a cooperative game the consequence is a coordination failure rather than a divergence: when the payoff for the best joint action is high but mismatching on it is heavily penalised, the penalties arrive during the other agent's exploration and are charged to the first agent's own action, which can drive both towards a safe but inferior agreement. The coordination and negotiation mechanisms of Experiments 6 and 7 were designed by hand; here the agents must arrive at one by themselves.

Algorithm

Algorithm 1 — Single-agent Q-learning

1. Build a 5 by 5 grid world with the start at one corner and a goal at the opposite one rewarding +10, a pit cell rewarding −10, both terminal, and a reward of −1 for every other move; a move off the grid leaves the agent where it is. Initialise Q(s, a) for all four moves in every cell, and set the learning rate, the discount factor, the exploration rate and the number of episodes.
2. In each episode start at the corner and repeat until a terminal cell is reached or a step limit expires: choose a random action with probability ε and otherwise the one with the highest Q, take it, observe the reward and the next cell, and move Q(s, a) towards the reward plus the discounted best Q of that next cell. Record the return of every episode, and at the end print the greedy policy as a grid, the number of steps it needs from the start and the episode at which it first became optimal.

Algorithm 2 — Two independent Q-learners in one game

1. Define a cooperative game in which both agents choose one of three actions and both receive the same payoff: 10 for either of the two matching optimal choices, 2 for a safe middle choice, a large negative penalty k when the two optimal choices are mismatched, and 0 otherwise. Give each agent one estimate per action of its own, with no state.
2. Repeat for many rounds: each agent picks its action ε-greedily from its own estimates, both receive the common payoff, and each moves the estimate of the action it chose towards that payoff. Report the final estimates of both agents, the pair of actions they settle on, and whether that pair is the optimal joint action, the safe one or a mismatch.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the greedy policy grid with its path length and the episode at which it first became optimal, the mean return early and late in training, the payoff matrix of the game, and the final estimates and chosen actions of both learners.

Observations

Comment on the difference between the episode at which the greedy policy became optimal and the return the agent was still earning at that point, on why two actions share the highest value at the start cell, on what the exploration of the second agent does to the first agent’s estimate for its own best action, and on whether the pair of actions the learners settle on is an equilibrium of the game they are playing.

Exercises

1. Sweep the exploration rate over 0, 0.05, 0.1 and 0.3 and the learning rate over 0.1 and 0.5, repeating each setting with ten seeds, and report how many runs end with an optimal greedy policy, the median episode at which it first became optimal and the mean return over the last fifty episodes. Then repeat the zero exploration setting with all Q values initialised to −20 instead of 0, and explain the difference in the result.
2. Sweep the penalty k over 0, −20, −50 and −100 with fifty runs each, reporting how often the two learners end at the optimal joint action, at the safe one and mismatched. Then replace the averaging update with one that sets the estimate of an action to the best payoff ever received for it, repeat the sweep, and explain why the outcome changes.